Durgesh Dutt Sinha

AIML Engineer & Full-Stack Developer

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PROFESSIONAL SUMMARY

Results-driven **AI/ML Engineer & Full-Stack Architect** with deep expertise in PyTorch, computer vision, real-time multimodal systems, and modern Next.js/FastAPI production architectures. Proven record architecting full-duplex WebSocket voice assistants, unsupervised customer segmentation pipelines, and deep learning vision systems. Master of Computer Applications (AIML) candidate with experience deploying containerized AI pipelines and driving measurable performance optimizations.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C/C++, HTML5, CSS3

AI & Machine Learning: PyTorch, Scikit-Learn, Computer Vision (MediaPipe, OpenCV, CNN), Whisper STT, Gemini API, Gemini 2.0 Flash, RAG, Pandas, NumPy, K-Means, DBSCAN

Full-Stack & Backend: Next.js, React, FastAPI, Node.js, WebSockets, RESTful APIs, Tailwind CSS, Streamlit

Cloud, Database & Tools: PostgreSQL (Neon), Drizzle ORM, Docker, Cloudflare Pages, Git & GitHub, Linux, AWS Educate

FEATURED PROJECTS

J.A.R.V.I.S. — Real-Time Multimodal AI Voice Assistant — GitHub • Live Demo

Tech Stack: FastAPI • Python • WebSockets • React • TypeScript • Whisper STT • Gemini 2.0 Flash • Edge-TTS

- **Full-Duplex Voice Engine:** Engineered a full-duplex voice assistant streaming 16kHz audio over bi-directional WebSockets between React frontend and FastAPI backend, achieving <450ms turnaround latency.
- **Multimodal AI Integration:** Combined Silero Voice Activity Detection (VAD) with Whisper STT and Gemini 2.0 Flash streaming API, paired with Edge-TTS for seamless real-time voice interruptions.
- **Fault-Tolerant Streaming:** Developed real-time audio wave telemetry and exponential backoff WebSocket reconnection state machines, ensuring zero-crash streaming resilience across unstable networks.

RoleRadar — AI Resume Analyzer & Career Intelligence Platform — GitHub • Live Demo

Tech Stack: Next.js • React • TypeScript • Drizzle ORM • PostgreSQL (Neon) • Tailwind CSS • pdf-parse

- **Resume Parsing Engine:** Architected full-stack resume analysis engine parsing PDF/DOCX files and scoring candidate alignment against 16 canonical engineering role taxonomies in under 1.2 seconds.
- **Google XYZ Scoring:** Implemented Google XYZ-formula scoring, ATS keyword density analyzers, and actionable gap audits; persisted candidate benchmark reports to Neon PostgreSQL via Drizzle ORM.
- **Edge UI Architecture:** Designed a responsive, accessible user interface with real-time scoring feedback loops and zero-SSR cold starts on Cloudflare Pages edge runtime.

MarketMatch-AI — Customer Segmentation & Lookalike Analytics Studio — GitHub • Live Demo

Tech Stack: Python 3.11 • Scikit-Learn • Streamlit • Plotly 3D • Pandas & NumPy • Joblib

- **Unsupervised ML Pipeline:** Developed unsupervised machine learning pipeline clustering 4,300+ e-commerce transaction records using RFM (Recency, Frequency, Monetary) feature extraction.
- **Algorithm Benchmarking:** Benchmarked K-Means clustering (Silhouette Score: 0.64, K=5) against DBSCAN for anomaly suppression; visualized high-dimensional clusters with interactive 3D Plotly projections.
- **Production Artifact Inference:** Exported serialized model artifacts via Joblib, enabling instantaneous lookalike cohort scoring and dynamic cohort CSV exports in an interactive Streamlit UI.

EXPERIENCE & FELLOWSHIPS

AI Program Fellow — UNLOX® — Jan 2026 – Present • Remote

- **Autonomous Agent Research:** Evaluated autonomous agent architectures, function-calling workflows, and LLM guardrails under senior technical mentorship.
- **Dockerized ML Pipelines:** Containerized multi-service machine learning pipelines with Docker, reducing environment setup drift and accelerating model prototyping cycles by 40%.

AI Cohort Member & Prototype Builder — Be10x — Oct 2025 – Dec 2025 • India

- **Prompt-Chaining Workflows:** Designed prompt-chaining architectures and agentic automation workflows, optimizing token consumption across multi-step LLM tasks.
- **AI Automation Prototypes:** Built functional proof-of-concept AI productivity automations, integrating structured JSON output schemas and multi-turn conversational memory.

Web Development Intern — InAmigos Foundation (IAF) — Jun 2024 – Aug 2024 • Remote

- **Accessible Web Interfaces:** Developed and maintained responsive web interfaces for non-profit initiatives using modern JavaScript and semantic HTML5/CSS3, improving mobile accessibility to WCAG 2.1 AA standards across 5,000+ monthly visitors.

EDUCATION

Master of Computer Applications (AIML) — Jul 2025 – 2027 (Expected)
Sri Balaji University, Pune
Specialization: Artificial Intelligence & Machine Learning

Bachelor of Computer Applications — Aug 2022 – Apr 2025
Sri Balaji University, Pune
Focus: AI, System Architecture, Prompt Engineering

CERTIFICATIONS & HONORS

- **AWS Educate Emerging Talent Community Member**
  Amazon Web Services (AWS)
- **AWS Educate Machine Learning Foundations**
  Amazon Web Services (AWS)